

Training-inference data skew: While drift happens over time, data skew is a structural discrepancy. It is often caused by technical debt or architectural flaws.

- **Feature engineering mismatch:** If your training pipeline uses a Python library to calculate a ‘7-day rolling average’ and your production C++ backend uses a slightly different windowing logic, you have skew.
- **Data leakage:** A classic mistake where a feature that is only available *after* the event (e.g., ‘total call duration’ for a churn model) is accidentally used in training.

Common drift detection methods

When monitoring models in production, we can’t rely on gut feel or visual inspection alone. Small shifts may look harmless but compound over time, while large shifts may be obvious but hard to justify without evidence. What we need is a systematic way to answer one question: Does today’s data behave differently from what the model was trained on?

Drift detection methods give us objective signals to answer this by comparing the current data distribution to a reference (training or baseline) distribution.

Distribution comparison (statistical distance): At a high level, these methods ask: How different are two distributions?

Numerical features: For continuous or numerical variables (e.g., price, latency, session duration), the Kolmogorov–Smirnov (K-S) Test is commonly used, which compares the shape of two distributions, not just their averages. It is non-parametric, meaning it makes no assumptions about normality. The output is a single distance score representing the maximum separation between the two distributions. If this distance exceeds a threshold, we conclude that the feature has statistically drifted.

Categorical features: For discrete features such as country, device type, or product category, numerical distance tests are not applicable. We can either use the Chi-Squared Test or the Jensen-Shannon Divergence. The Chi-Squared Test checks whether category frequencies have changed more than expected by chance. The Jensen-Shannon Divergence (JSD) measures how similar two probability distributions are, in a bounded and symmetric way.

Population Stability Index (PSI): While statistical tests tell us whether a change is significant, teams often want a single, easy-to-interpret score. This is where PSI is widely used in industry. However, its thresholds should be treated as a guideline.

The commonly used ranges are:

- $PSI < 0.1$ (no change)
- $0.1-0.25$ (moderate drift)
- 0.25 (significant drift)

In high-risk systems (fraud detection, medical models, credit underwriting), even small PSI values may warrant investigation. In noisy consumer data (clickstream,

marketing), larger PSI values may be acceptable. The correct threshold is ultimately a business decision, not just a statistical one. It should reflect how costly it is for the model to be wrong.

Tools and frameworks

The MLOps ecosystem has matured, offering tools that cater to different scales and needs.

Tool	Focus	Why use it?
Evidently	Visual analysis	Generates beautiful, interactive HTML reports. Perfect for ‘Debug Days’ where data scientists need to see where the drift is happening.
WhyLogs	Data profiling	Uses ‘Data sketching’ to create tiny, mergeable summaries of data. Essential for high-volume pipelines (Big Data), where moving raw data is too expensive.
MLflow	Lifecycle tracking	While primarily for experiment tracking, its integration with monitoring tools allows you to link drift alerts directly to the specific model version and its training parameters.
Great Expectations	Data quality	Focuses on ‘Unit Testing for Data’. It ensures that incoming data meets specific schemas and distributions before it ever hits the model.

Hands-on example: Identifying drift in housing data

Let’s look at a practical scenario. We will use the California Housing dataset from *sklearn*, train a model on one subset and then ‘corrupt’ a second subset to simulate an economic shift (drastic increase in house prices and decrease in local income).